

✓ 1. Resolve the following system using the Cramer's Rule.

$$0.14x_1 - 0.1x_2 - 0.2x_3 = 7.85$$

$$0.10x_1 + 7x_2 - 0.3x_3 = -19.3$$

$$0.30x_1 - 0.2x_2 + 10x_3 = 71.4$$

**Sol<sup>n</sup>:** Let us write these equations in the form  $AX=B$

$$\begin{bmatrix} .14 & -.1 & -.2 \\ .1 & 7 & -.3 \\ .3 & -.2 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7.85 \\ -19.3 \\ 71.4 \end{bmatrix}$$

$$D=|A| = \begin{vmatrix} .14 & -.1 & -.2 \\ .1 & 7 & -.3 \\ .3 & -.2 & 10 \end{vmatrix} = 10.3246$$

$$D_{x_1} = \begin{vmatrix} 7.85 & -.1 & -.2 \\ -19.3 & 7 & -.3 \\ 71.4 & -.2 & 10 \end{vmatrix} = 631.059$$

$$D_{x_2} = \begin{vmatrix} .14 & 7.85 & -.2 \\ .1 & -19.3 & -.3 \\ .3 & 71.4 & 10 \end{vmatrix} = -35.1637$$

$$D_{x_3} = \begin{vmatrix} .14 & -.1 & 7.85 \\ .1 & 7 & -19.3 \\ .3 & -.2 & 71.4 \end{vmatrix} = 54.0826$$

$$x_1 = D_{x_1}/D = 631.059/10.3246 \approx 22.135$$

$$x_2 = D_{x_2}/D = -35.1637/10.3246 \approx -3.405$$

$$x_3 = D_{x_3}/D = 54.0826/10.3246 \approx 5.238$$

✓ 2. Use the Gauss-Elimination technique to resolve the following system.

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85$$

$$0.10x_1 + 7x_2 - 0.3x_3 = -19.3$$

$$x_1 - 0.2x_2 + 10x_3 = 71.4$$

**Sol<sup>n</sup>:**

Write the augmented matrix for the system:

$$\begin{bmatrix} 3 & -0.1 & -0.2 & | & 7.85 \end{bmatrix}$$

$$\begin{bmatrix} 0.1 & 7 & -0.3 & | & -19.3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -0.2 & 10 & | & 71.4 \end{bmatrix}$$

Perform row operations to eliminate the coefficients below the leading diagonal element in the first column:

Multiply the first row by 1/3 to make the leading coefficient 1:

$$\begin{bmatrix} 1 & -0.033333 & -0.066667 & | & 2.6167 \\ 0.1 & 7 & -0.3 & | & -19.3 \\ 1 & -0.2 & 10 & | & 71.4 \end{bmatrix}$$

Subtract 0.1 times the first row from the second row to make the second element in the first column 0:

$$\begin{bmatrix} 1 & -0.033333 & -0.066667 & | & 2.6167 \\ 0 & 7.003333 & -0.293333 & | & -21.583 \\ 1 & -0.2 & 10 & | & 71.4 \end{bmatrix}$$

Subtract the first row from the third row to make the third element in the first column 0:

$$\begin{bmatrix} 1 & -0.033333 & -0.066667 & | & 2.6167 \\ 0 & 7.003333 & -0.293333 & | & -21.583 \\ 0 & -0.166667 & 10.066667 & | & 68.783 \end{bmatrix}$$

Now, normalize the second row by dividing it by 7.003333 (the leading coefficient in the second row):

$$\begin{bmatrix} 1 & -0.033333 & -0.066667 & | & 2.6167 \\ 0 & 1 & -0.041879 & | & -3.0811 \\ 0 & -0.166667 & 10.066667 & | & 68.783 \end{bmatrix}$$

Add 0.166667 times the second row to the third row to make the third element in the second column 0:

$$\begin{bmatrix} 1 & -0.033333 & -0.066667 & | & 2.6167 \\ 0 & 1 & -0.041879 & | & -3.0811 \\ 0 & 0 & 10.267144 & | & 69.9422 \end{bmatrix}$$

Normalize the third row by dividing it by 10.267144 (the leading coefficient in the third row):

$$\begin{bmatrix} 1 & -0.033333 & -0.066667 & | & 2.6167 \\ 0 & 1 & -0.041879 & | & -3.0811 \\ 0 & 0 & 1 & | & 6.8077 \end{bmatrix}$$

Back-substitute to find the values of x:

From the third row, we have:

$$x_3 = 6.8077$$

From the second row, we have:

$$x_2 - 0.041879x_3 = -3.0811$$

$$x_2 - 0.041879(6.8077) = -3.0811$$

$$x_2 \approx -3.0811 + 0.2836$$

$$x_2 \approx -2.50$$

From the first row, we have:

$$x_1 - 0.033333x_2 - 0.066667x_3 = 2.6167$$

$$x_1 - 0.033333(-2.7975) - 0.066667(6.8077) = 2.6167$$

$$x_1 \approx 2.6167 + 0.09325 + 0.4538$$

$$x_1 \approx 3.0$$

So, the solution to the system is:

$$x_1 \approx 3.0$$

$$x_2 \approx -2.50$$

$$x_3 \approx 6.8077$$

### 3. Apply the Factorization process to locate the root of the following system:

$$x_1 + x_2 - x_3 = 2$$

$$2x_1 + 3x_2 + 5x_3 = -3$$

$$3x_1 + 2x_2 - 3x_3 = 6$$

○ **Sol<sup>n</sup>:** Rules or technique  $\rightarrow AX=B \Rightarrow LUX=B \Rightarrow LY=B$  then  $UX=Y$

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 2 & 3 & 5 \\ 3 & 2 & -3 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, B = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}$$

A matrix can be written as the product of a lower triangular matrix and an upper triangular matrix

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 2 & 3 & 5 \\ 3 & 2 & -3 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

Comparing the values,

$$u_{11}=1, u_{12}=-1, u_{13}=-1, l_{21}u_{11}=2, \text{ so that } l_{21}=2$$

Similarly, we can find out another unknown variable,

$$AX=B \Rightarrow LUX=B$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 7 \\ 0 & 0 & 7 \end{bmatrix} X = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}$$

Let,  $UX=Y \Rightarrow LY=B$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}$$

From above matrix we find,

$$\checkmark \quad y_1 = 2$$

$$\checkmark \quad 2y_1 + y_2 = -3$$

$$\Rightarrow y_2 = -7$$

$$\checkmark \quad 3y_1 - y_2 + y_3 = 6 \Rightarrow y_3 = -7$$

Again,  $UX=Y$

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 7 \\ 0 & 0 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -7 \\ -7 \end{bmatrix}$$

From above matrix we find,

$$\checkmark \quad 7x_3 = -7$$

$$x_3 = -1$$

$$\checkmark \quad x_2 + 7x_3 = -7$$

$$\Rightarrow x_2 = 0$$

$$\checkmark \quad x_1 + x_2 - x_3 = 2$$

$$\Rightarrow x_1 = 1$$